



Classification of bacterial images obtained optically using some pre-trained models

Lizette Nange Chia^{a,d}, Thomas Tamo Tatietse^b, Gemma Piella^c, Jimmi Hervé Talla Mbé^{a,*}

^a Department of Physics, Research Unit of Condensed Matter, Electronics and Signal Processing, Faculty of Sciences, University of Dschang, P.O. Box 67, Cameroon

^b Department of Civil Engineering, National Advanced School of Engineering, University of Yaoundé I, Yaoundé, P.O. Box 510, Cameroon

^c Department of Engineering, Universitat Pompeu Fabra, Barcelona 08018, Spain

^d African Institute of Mathematical Sciences (AIMS), Limbe, P.O. Box 608, Cameroon

ARTICLE INFO

Keywords:

Water-borne bacteria
Convolutional neural networks (CNNs)
Image classification
ResNet

ABSTRACT

Rapid and accurate detection of water-borne bacteria is critical for safeguarding public health and preventing the spread of infections. Conventional bacterial identification techniques are often time-consuming, resource-intensive, and require specialized personnel, limiting their suitability for automated water-quality monitoring. This study presents an optical detection and classification framework that uses some pre-trained architectures to automatically classify optically acquired images of water-borne bacterial pathogens. The system distinguishes among four classes: *Escherichia coli* (*E. coli*), Fecal streptococci (*Strept*), co-occurrence of both bacteria (*Both*), and safe water (*None*). Experimental evaluation demonstrates strong classification performance, with ResNet-50, ResNet-152, and EfficientNet-B7 achieving accuracies of 94.15%, 94.45%, and 95.54%, respectively. Except DenseNet-201, which yields the worst results (accuracy of 77.49%), the corresponding precision, recall, and F1-scores of the other pre-trained models exceed 94%. Furthermore, an analysis of the ROC (Receiver Operating Characteristic) curve reveals that the Area Under the Curve (AUC) exceeds 98% for each bacterial class, thereby demonstrating strong discriminative performance. The results highlight the potential of transfer learning-based convolutional neural networks for accurate, rapid, and cost-effective bacterial detection, emphasizing their promise for scalable and automated water-quality monitoring systems. This approach represents a step toward improved monitoring tools aligned with sustainable water, sanitation, and hygiene (WASH) objectives.

1. Introduction

Ensuring water safety is a critical public health challenge, as contamination by pathogenic bacteria such as rod-shaped *Escherichia coli* and spherical Fecal streptococci poses serious health risks (Lekeufack et al., 2012; Ntangmo Tsafack et al., 2022). Conventional microbiological methods for detecting and classifying bacteria are labor-intensive, time-consuming, and require specialized expertise, making them less suitable for rapid or large-scale monitoring (Nurliyana et al., 2018; Tang et al., 2024). To overcome these limitations, automated image-based machine learning classification methods have been explored, including Support Vector Machines (SVMs), k-Nearest Neighbor, and Random Forest (Dinesh et al., 2024; Murugan et al., 2019). However, these classifiers typically rely on handcrafted features, which limit their scalability and generalization when applied to high-dimensional and complex image data (Ahmed et al., 2023; Dargan

et al., 2020; Jan et al., 2019; Wang et al., 2021). The availability of large annotated datasets such as ImageNet has driven the development of deep learning methods that automatically extract hierarchical features directly from raw data, overcoming many limitations of conventional approaches (Krizhevsky et al., 2012; Muhammed et al., 2017; Ojha et al., 2017).

In microbiology, deep learning methods, particularly convolutional neural network (CNN)-based models, have demonstrated substantial improvements in automating bacterial classification, providing faster and more accurate alternatives to traditional methods (Poomritigul et al., 2022; Shaily and Kala, 2020; Talo, 2019; Tanko et al., 2024). By automatically extracting spatial and morphological features, CNNs reduce the dependence on handcrafted feature engineering. CNNs have recently gained attention for the classification of bacteria from optically acquired images (Chia et al., 2025). In that study, images were obtained using the water-drop method and prepared for subsequent classification.

* Corresponding author.

E-mail addresses: gemma.piella@upf.edu (G. Piella), jhtallam@yahoo.fr (J.H. Talla Mbé).

<https://doi.org/10.1016/j.mimet.2026.107572>

Received 12 April 2026; Received in revised form 3 June 2026; Accepted 4 June 2026

Available online 10 June 2026

0167-7012/© 2026 Elsevier B.V. All rights reserved, including those for text and data mining, AI training, and similar technologies.

A shallow four-layer CNN was used to classify four bacterial species achieving an accuracy of 89.8% after 50 epochs (Chia et al., 2025). While these results are promising, shallow networks lack sufficient depth to model complex spatial patterns effectively, a limitation reflected in their lower classification performance compared to deeper architectures. To address these limitations, deeper and more advanced networks such as DenseNet, ResNet and EfficientNet can be used (He et al., 2015; Simonyan and Zisserman, 2014; Szegedy et al., 2015). These models have proven especially effective in mitigating gradient degradation and stabilizing training in very deep networks (Bhutipia et al., 2024; Borawar and Kaur, 2023; Gencer et al., 2024; He et al., 2015; Zhang and Gao, 2022). Building on these advances, this study investigates the application of transfer learning to classify water-borne bacterial images using optically acquired water-drop images. More specifically, we evaluate and compare the ResNet-50, ResNet-152, DenseNet-201, and EfficientNet-B7 architectures in this application domain. To the best of our knowledge, a systematic analysis of deep pre-trained networks for multi-class classification of water-borne bacteria in water-drop imagery has not been previously reported. In this regard, this study examines the application of transfer learning-based classification of bacterial images obtained by using the water-drop method. Our results demonstrate that transfer learning enables more robust feature extraction and improved classification performance compared to shallow CNN approaches, highlighting its suitability for accurate and scalable bacterial monitoring. Even if EfficientNet is known to perform better than several pre-trained models, our attention will be paid to Resnet variants for their simplicity and faster training, which make them easier to deploy in real time applications.

The remainder of this paper is structured as follows: Section 2 describes the dataset and the transfer learning models; Section 3 presents experimental results and comparative analyses; and Section 4 concludes with key findings and future directions.

2. Materials and methods

2.1. Dataset

The image dataset was acquired optically using the water-drop method illustrated in Fig. 1, inspired by the approach proposed by Planinsic (Planinsic, 2001). In this setup, a syringe is used to form a water droplet at its nozzle end, positioned at a fixed distance from a screen. A green laser beam with a wavelength of 532 nm is directed through the centre of the water droplet, perpendicular to the screen. This magnifies the shadow images of the bacteria suspended in the sample. The entire setup is enclosed in a dark box to prevent contamination from airborne microorganisms and interference from ambient light. The obtained images are categorized into four classes (Chia et al., 2025; Lekeufack et al., 2012; Planinsic, 2001): images containing only

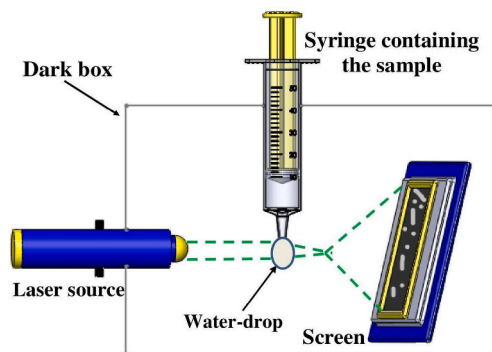


Fig. 1. 3D schematic illustration of the optically acquired images of water-borne bacterial pathogens, showing the path of the laser beam, the water droplet, and the projection screen.

Escherichia coli (labeled *E. coli*); images containing only fecal streptococci (labeled *Strept*); images where these two water-borne bacteria are both present (labeled *Both*); and those of non-contaminated water (labeled *None*) as shown in Fig. 2. The dataset is unbalanced, with the following class distribution: *E. coli* (551 images), *Strept* (1189 images), *Both* (1172 images), and *None* (2138 images). In total, the database comprises 5050 images, which are split into training (80%) and testing (20%) subsets. As the raw images vary in spatial resolution, all images are resized to 180×180 pixels. This reduces computational complexity while preserving morphological features necessary for water-borne bacterial discrimination (Al-Ani et al., 2025). To further improve model generalization, data augmentation techniques including random flipping, rotation, and zooming are applied during training.

2.2. Transfer learning and model adaptation

Pre-trained ResNet-50, ResNet-152, DenseNet-201, and EfficientNet-B7 models are employed here for bacterial image classification using transfer learning. ResNet-50 and ResNet-152 are deep residual convolutional neural networks composed of stacked residual blocks with skip connections, enabling effective training of very deep architectures (50 and 152 layers, respectively) by improving gradient flow and learning hierarchical feature representations. ResNet-50 strikes a balance between training efficiency and the ability to capture detailed features, making it well-suited for practical experimentation, whereas the significantly deeper ResNet-152 can extract highly complex hierarchical features. This allows us to evaluate the upper limits of classification performance and feature discrimination on our dataset to adapt to bacterial-specific patterns. DenseNet-201 is a densely connected convolutional network in which each layer receives feature maps from all preceding layers, promoting feature reuse and improving gradient propagation, resulting in a more parameter-efficient architecture. The DenseNet-201 is used for its performance on four object recognition benchmark datasets (Lu et al., 2021; Ait Amou et al., 2022). EfficientNet-B7 is a convolutional neural network that applies compound scaling to jointly optimize depth, width, and input resolution, and uses Mobile Inverted Bottleneck Convolutions (MBConv) with Squeeze-and-Excitation (SE) attention modules to enhance feature representation

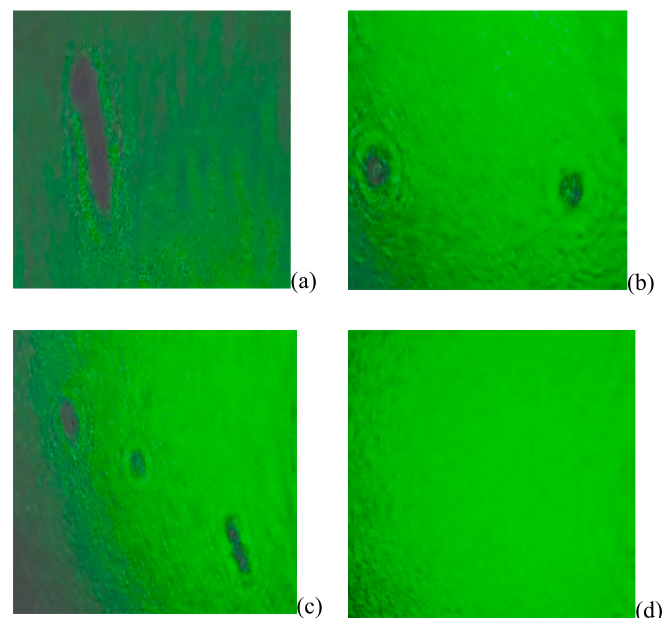


Fig. 2. Sample images from the water-drop optical acquisition dataset illustrating the four classes: (a) *Escherichia coli* only (*E. coli*), (b) Fecal streptococci only (*Strept*), (c) co-presence of both bacteria (*Both*), and (d) non-contaminated water (*None*).

efficiency. The EfficientNet-B7 is also used thanks to its compound scaling approach, efficient building blocks and attention mechanisms (Sharma, 2025; Khushi et al., 2023). The convolutional and pooling layers of the pre-trained networks, initialized with ImageNet weights, serve as fixed feature extractors during initial training. During training, all model layers are frozen, serving as a fixed feature extractor for our bacterial classification task. This strategy is particularly suited to limited datasets, as full fine-tuning of all layers may cause overfitting (Showkat and Qureshi, 2022). The original fully connected layers are replaced with a custom classifier of four dense layers (256, 128, 64, and 32 neurons) with ReLU activations and dropout regularization (rate 0.2), resulting in 567,908, 535,140, and 698,980 trainable parameters for the ResNets, DenseNet-201, and EfficientNet-B7, respectively (See Table 1). The final layer consists of four units with softmax activation for multi-class prediction. Batch normalization layers are incorporated to stabilize gradients and accelerate convergence (Amthor et al., 2016; Santurkar et al., 2018; Showkat and Qureshi, 2022). We use the Adam optimizer to improve the speed and stability of the model's convergence during training. For our four-class classification problem, in which each sample is assigned to exactly one of four independent classes, we use categorical cross-entropy as the loss function. Early stopping is applied to halt training when test performance plateaus.

2.3. Performance evaluation

Model performance is evaluated using standard classification metrics such as accuracy which is calculated as the proportion of correctly classified samples across all classes. Precision, recall, and F1-score are computed using a one-vs-rest scheme for each class and reported as weighted-averaged values, where each class contributes proportionally to its number of samples in the dataset. Confusion matrices are also used to visualize the distribution of correct and incorrect predictions across the four bacterial classes. In addition, we use the receiver operating characteristic (ROC) curve, via its area under the curve (AUC). The ROC curve is constructed from the true positive rate (TPR) and false positive rate (FPR) across different classification thresholds, providing a comprehensive measure of model performance over all operating points. The mathematical expressions of these metrics yield (Raja et al., 2026):

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \text{TPR} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{F1 score} = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

$$\text{FPR} = \frac{FP}{FP + TN} \quad (5)$$

Table 1
Architectural specifications of the pre-trained deep learning models.

Model	Backbone architecture	Key characteristics	Total parameters	Trainable parameters
ResNet-50	Residual CNN (50 layers)	Residual skip connections	24,155,620	567,908
ResNet-152	Residual CNN (152 layers)	Deeper hierarchical features	58,938,852	567,908
DenseNet-201	Dense CNN (201 layers)	Dense feature reuse	7,343,268	535,140
EfficientNet-B7	Compound-scaled CNN	MBConv + SE blocks	64,796,667	698,980

$$\text{Categorical cross entropy} = -\frac{1}{n} \sum_{i=1}^n \sum_{j=1}^C y_{ij} \log(p_{ij}) \quad (6)$$

where y_{ij} denotes the ground-truth label (one-hot encoded) for sample i and class j , p_{ij} is the probability that sample i is of class j , n is the number of samples, and C is the number of classes. TP , TN , FP , and FN represent true positives, true negatives, false positives, and false negatives, respectively.

3. Results

3.1. Overall performance

The classification performance of the proposed pre-trained ResNet architectures is evaluated in terms of the training and testing accuracy curves (Figs. 3, 4), precision, recall, and F1 score (Fig. 5). As illustrated in Fig. 3, the ResNet-50 model exhibits a progressive improvement in training accuracy, starting at approximately 65.91% and reaching 97.07% by 100 epochs, corresponding to a misclassification rate of about 3%. The testing accuracy follows a similar trend, beginning around 82.57% and stabilizing at 94.15%. The small gap between the training and testing accuracies ($\approx 2.92\%$) suggests strong generalization and minimal overfitting, indicating that the model effectively learns discriminative bacterial features. For ResNet-152 (Fig. 3), a comparable learning trend is observed. The models training accuracy increases from approximately 68.69% to 97.12%, while the testing accuracy is improved from 84.45% to 94.45%.

ResNet-50 presents a testing loss of 6.85%, whereas ResNet-152 achieves a slightly lower loss of 6.55%, indicating more efficient learning and stronger generalization capability. The consistency between both models highlights the robustness of residual learning across different network depths. Moreover, the deeper architecture of ResNet-152 contributes to enhancing feature representation and improving stability, confirming its suitability for optically acquired water-borne bacterial image classification tasks.

Table 2 summarizes the final training and testing accuracies for both models, highlighting their comparable performance. The Δ Accuracy values of 2.92% for ResNet-50 and 2.67% for ResNet-152 indicate minimal disparity between training and testing performance, reflecting strong generalization and stable learning in both architectures. The slight lower Δ Accuracy observed in ResNet-152 suggests improved consistency between training and testing outcomes, highlighting its enhanced ability to learn transferable features while maintaining robustness against overfitting.

The accuracy plots illustrate the learning behavior of both ResNet variants (Figs. 3 and 4). Initially, before the 10th epoch, early-stage irregularities occur in feature learning as both models exhibit minor

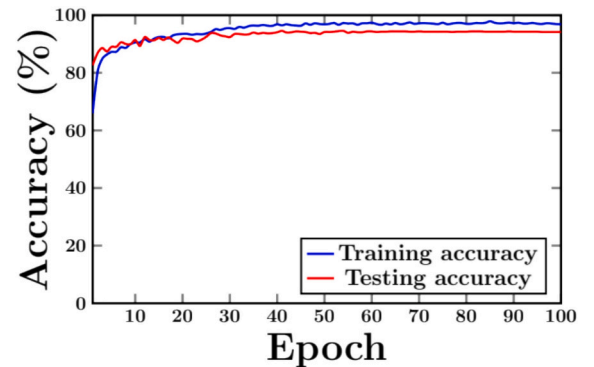


Fig. 3. Training and testing accuracy curves for ResNet-152 over 100 epochs, demonstrating how the model's accuracy improves and converges as it approaches the final epochs.

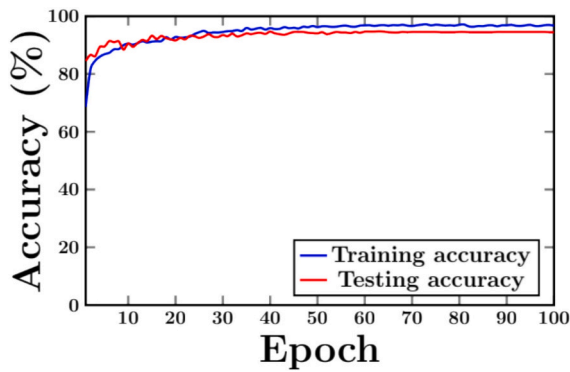


Fig. 4. Training and testing accuracy curves for ResNet-152 over 100 epochs, demonstrating how the model's accuracy improves and converges as it approaches the final epochs.

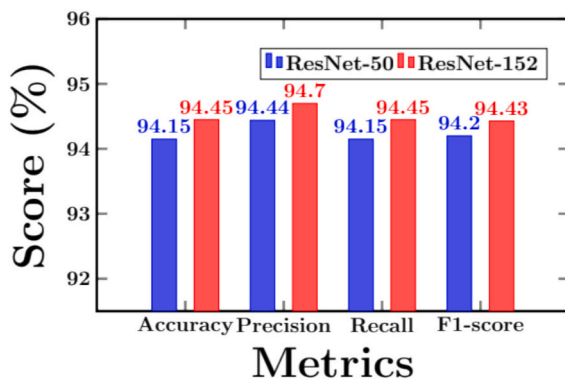


Fig. 5. Comparison of the weighted classification performance metrics for ResNet-50 and ResNet-152, trained for 100 epochs.

Table 2
Performance comparison of ResNet-50 and ResNet-152 models for 100 epochs.

Metric	ResNet-50	ResNet-152
Training Accuracy (%)	97.07	97.12
Testing Accuracy (%)	94.15	94.45
Δ Accuracy (%)	2.92	2.67

instances of underfitting, where the testing accuracy temporarily exceeds that of the training. Subsequently, the ResNet-50 model enters a transient phase between the 10th and 20th epochs, while the ResNet-152 model experiences a longer transition period extending from the 10th to the 30th epochs. During these intervals, both training and accuracies fluctuate considerably since both models are adjusting their internal representations. After the 20th (ResNet-50) and 30th (ResNet-152) epochs, both models demonstrate improved generalization and consistent performance on the database through stabilized accuracy. This stabilization also manifests with no significant change in both training and testing accuracies as the number of epochs increases.

Further comparative analyses on both ResNet variants employing precision, recall, and F1-score metrics are presented in Fig. 5. ResNet-50 demonstrates consistently high performance, reaching an accuracy of 94.15%, with precision, recall, and F1-score values of 94.44%, 94.15%, and 94.20%, respectively. ResNet-152 rather attains even stronger results, with an accuracy of 94.45%, precision of 94.70%, recall of 94.45%, and an F1-score of 94.43%. As illustrated in Fig. 5, this improvement, although numerically moderate, reflects the deeper model's enhanced capacity to capture fine-grained differences between classes while preserving the overall robustness observed in both

architectures. The close alignment among the four metrics further confirms that the two networks generalize reliably, with ResNet-152 offering an advantage in terms of overall classification. These results indicate that both architectures are well-suited for the bacterial classification task, with the deeper model providing a measurable performance benefit.

3.2. Performance per class

We now evaluate the model's performance per bacterial class using confusion matrices (Figs. 6 and 7), evaluation metrics in Table 3, and the ROC curves (Figs. 8 and 9).

Confusion matrices exhibit a strong diagonal dominance, confirming that the majority of classes are correctly classified across all bacterial categories. The *None* and *Strept* classes appear with the highest accuracies. Therefore, the model is robust in distinguishing clear spherical bacterial morphologies and background features. Besides, both ResNet-50 and ResNet-152 variants demonstrate identical performance for the *None* class, suggesting that both architectures effectively recognize background regions or class representing safe, bacteria-free water conditions. A relatively lower performance is observed for the *E. coli* class, although ResNet-50 achieves slightly better discrimination than ResNet-152. Specifically, ResNet-50 misclassifies approximately 11% of *E. coli* class as *Strept* class and 9.1% of *Both* class as *Strept* class, whereas ResNet-152 misclassifies 18% of *E. coli* as *Strept*, but reduces the confusion between *Both* and *Strept* to 5.9%. Such misclassification could originate from the low proportion of *E. coli* class compared to other classes.

To further quantify the classification performance observed in the confusion matrices and the comparison of weighted average metrics in Table 2, we present the precision, recall, and F1 score for ResNet-50 and ResNet-152, both of which were trained over 100 epochs (Table 3).

Both models demonstrate high precision across all classes, indicating reliable positive predictions. However, the *E. coli* class shows relatively lower precision in ResNet-50 (89.04%) compared to ResNet-152 (94.97%), which is consistent with the higher rate of false positives observed in the confusion matrix. In terms of recall, the *Strept* class achieves the highest values for both models, with ResNet-152 outperforming ResNet-50, indicating strong sensitivity to this class. In contrast, the *E. coli* class records the lowest recall for both models, suggesting that a significant proportion of *E. coli* samples are not correctly classified, possibly due to class imbalance. The *None* class achieves the highest performance consistently across all metrics, exceeding 98% for both models, reflecting its dominance in the dataset. The F1-score indicates that ResNet-50 maintains more balanced performance across all classes, whereas ResNet-152 shows greater variability, particularly for the weakly represented *E. coli* class.

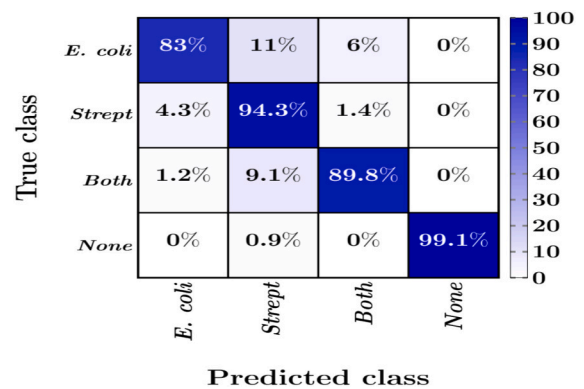


Fig. 6. Confusion matrix for ResNet-50, trained over 100 epochs, with cell values showing sample proportion and color intensity indicating classification concentration.

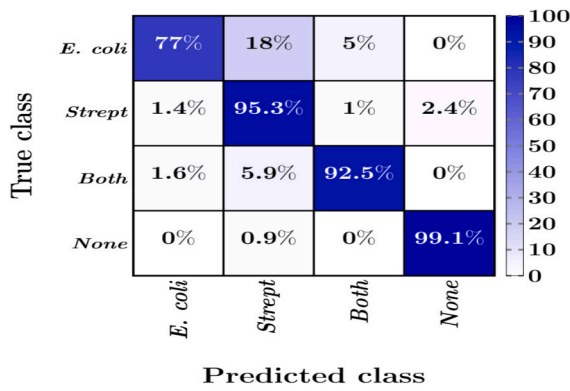


Fig. 7. Confusion matrix for ResNet-152 at 100 epochs, with cell values showing sample proportion and color intensity indicating classification concentration.

Table 3

Per-class classification metrics (precision, recall, and F1-score) for ResNet-50 and ResNet-152 trained for 100 epochs.

		Precision	Recall	F1-score
		(%)	(%)	(%)
ResNet-50	<i>E.coli</i>	89.04	82.00	85.38
	<i>Strept</i>	84.85	95.84	90.00
	<i>None</i>	91.54	81.58	92.46
	<i>Both</i>	99.77	98.88	99.32
ResNet-152	<i>E. coli</i>	94.97	79.00	85.24
	<i>Strept</i>	87.59	97.73	92.38
	<i>None</i>	97.31	94.30	95.80
	<i>Both</i>	99.33	99.33	99.33

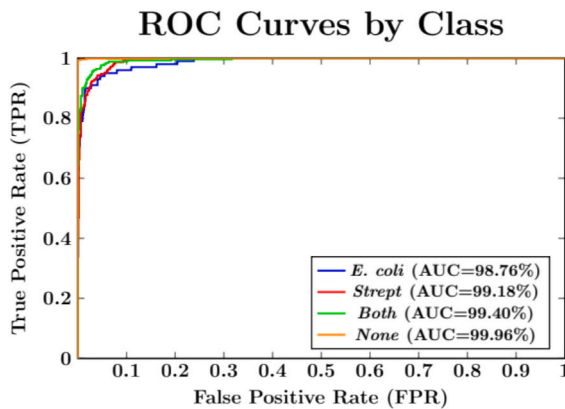


Fig. 8. ROC curves with ResNet-50 trained over 100 epochs, illustrating the trade-off between the true positive rate (TPR) versus the false positive rate (FPR) for different bacterial classes.

The receiver operating characteristic (ROC) and their corresponding area under the curve (AUC) values for both architectures are shown in Figs. 8 and 9. Both ResNet-50 and ResNet-152 achieves AUCs exceeding 98% across all classes, indicating excellent discriminative capability in distinguishing bacterial species and the negative class. The marginally higher AUCs for ResNet-152, particularly in the *Strept*, *None*, and *Both* classes, demonstrate that its deeper residual architecture facilitates improved feature extraction and class separability, especially for visually overlapping bacterial structures. These high AUC scores demonstrate the models' robustness and reliability in bacterial classification. It is also observed that the *None* class exhibits the highest AUC for both variants with a relatively small difference. In contrast, *E. coli* has the smallest AUC for both variants, with ResNet-50 portraying a higher

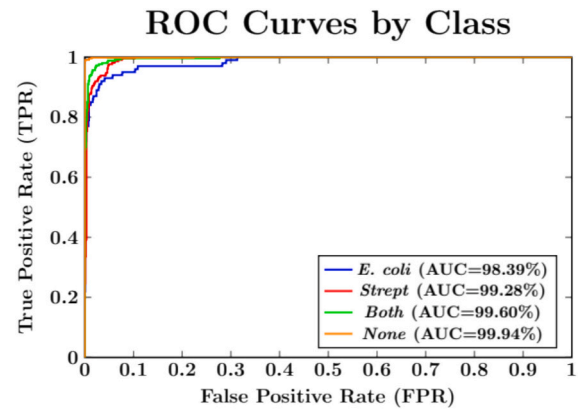


Fig. 9. ROC curves for ResNet-152 trained over 100 epochs, illustrating the trade-off between the true positive rate (TPR) versus the false positive rate (FPR) for different bacterial classes.

AUC. These results confirm those obtained in the confusion matrices.

3.3. Comparative study of pre-trained models

Table 4 presents a detailed comparison of the models' inference times, showing that ResNet-50 requires less time than ResNet-152. Table 4 also highlights the performance of the ResNet models relative to DenseNet-201 and EfficientNet-B7. DenseNet-201 yields the poorest results across all these performance metrics and exhibits a very high inference time per image compared to ResNet-50, albeit close to that of ResNet-152 and EfficientNet-B7. This suggests that, in this context of bacterial classification using the optically acquired water-drop method, dense connections do not significantly improve performance compared to residual connections. Conversely, it is observed that EfficientNet-B7 outperforms the ResNet models on nearly all metrics, thereby demonstrating effective discrimination of bacterial morphology. For instance, its accuracy is 1.39% and 1.09% higher than that of ResNet-50 and ResNet-152, respectively. Nevertheless, it is worth noting that its inference time per image is 9.56 ms and 2.02 ms longer than that of ResNet-50 and ResNet-152, respectively, which could translate into a non-negligible total inference time. It is also important to note that DenseNet-201 and EfficientNet-B7 exhibit trends similar to those of ResNet-50 and ResNet-152 regarding accuracy and ROC curves (Figs. 3, 4, 8, 9), as well as respectively-lower and higher classification performance for the *E. coli* and *None* classes (Figs. 6, 7) (not shown).

4. Conclusion

In this paper, we have proven that ResNet variants can be combined with a simple optical water-drop bacterial image acquisition for accurately classifying different types of water-borne pathogens present in water. Two architectures, ResNet-50 and ResNet-152, have been particularly evaluated in more details within a transfer-learning

Table 4

Comparison of classification performances of ResNet models and those of DenseNet-201 and EfficientNet-B7 for 100 epochs. The same parameters are used for the models.

Model	Precision (%)	Recall (%)	F1-score (%)	Accuracy (%)	Inference time (per image)
ResNet-50	94.44	94.15	94.2	94.15	3.49 ms
ResNet-152	94.7	94.45	94.43	94.45	11.03 ms
DenseNet-201	76.13	77.62	76.36	77.62	11.28 ms
EfficientNet-B7	95.53	95.54	95.52	95.54	13.05 ms

framework. Both models have demonstrated high performance across several metrics such as accuracy, precision, Recall, F1-score, confusion matrix, and ROC. Indeed, good performances have been relatively achieved after a few epochs (~30).

In general, ResNet-152 consistently yields slight improvements over ResNet-50 (less than 1%) due to its greater depth, which allows more effective feature extraction and better discrimination between bacterial classes, particularly in mixed or visually complex samples. However, that small gap could not compensate for the higher computational resources required by ResNet-152 compared to ResNet-50. Nevertheless, results from those ResNet variants provide compelling evidence that ResNet-based transfer learning solutions can generalize effectively, even under constraints of limited training data and inherent variability in optical bacteria images. A comparison with other pre-trained models demonstrates that ResNet-50 and ResNet-152 achieve significantly better results than DenseNet-201, while coming very close to EfficientNet-B7 which requires meticulous hyperparameter tuning, consumes substantial computational resources, and currently has a smaller user community. However, the ResNet variants exhibit faster inference times. From an application perspective, the performances obtained are promising, demonstrating that these pre-trained networks are capable of distinguishing water contaminated solely by *Escherichia coli* (*E. coli*) from water containing only Fecal streptococci (*Strept*), from water infested with both bacteria (*Both*), as well as from safe water (*None*). Overall, this work establishes deep learning-assisted optical classification as a promising pathway for rapid, automated bacterial detection in water quality monitoring. Such systems could accelerate contamination assessment, reduce dependence on slow culture-based methods, and enhance early-warning capabilities for water-borne disease prevention. We envisage forthcoming works to deal with other alternative strategies and novel machine learning approaches (Li et al., 2026; Liedji et al., 2022; Liedji et al., 2023; Wenkack Liedji et al., 2022).

CRedit authorship contribution statement

Lizette Nange Chia: Software, Methodology, Formal analysis, Conceptualization. **Thomas Tamo Tatietse:** Supervision, Writing – review & editing. **Gemma Piella:** Supervision, Writing – review & editing, Writing – original draft. **Jimmi Hervé Talla Mbé:** Supervision, Project administration, Data curation, Conceptualization, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

We would like to express our heartfelt gratitude to the Congolese Foundation for Biomedical 327 Research for their support. JHTM acknowledge Edmund Optics Educational awards for support.

Data availability

Data will be made available on request.

References

- Ahmed, S.F., Alam, M.S.B., Hassan, M., Rozbu, M.R., Ishtiaq, T., Rafa, N., Mofijur, M., Shawkat Ali, A., Gandomi, A.H., 2023. Deep learning modelling techniques: current progress, applications, advantages, and challenges. *Artif. Intell. Rev.* 56 (11), 13521–13617.
- Ait Amou, M., Xia, K., Kamhi, S., Mouhafid, M., 2022. A novel MRI diagnosis method for brain tumor classification based on CNN and Bayesian optimization. *Healthcare* 10 (3), 494. MDPI.
- Al-Ani, S., Guo, H., Fyfe, S., Long, Z., Donnaz, S., Kim, Y., 2025. Effect of training sample size, image resolution and epochs on filamentous and floc-forming bacteria classification using machine learning. *J. Environ. Manag.* 379, 124803.
- Amthor, M., Rodner, E., Denzler, J., 2016. Impatient dnns-deep neural networks with dynamic time budgets. *arXiv Preprint arXiv:1610.02850*.
- Bhuthia, R., Dave, S.A.K., Mallela, I.R., Mall, R., Das, A., Joshi, A., 2024. Optimizing web interfaces with AI-generated CSS through machine learning and deep learning. In: 2024 13th International Conference on System Modeling & Advancement in Research Trends (SMART), pp. 287–292.
- Borawar, L., Kaur, R., 2023. ResNet: Solving vanishing gradient in deep networks. *Proc. Int. Conf. Recent Trends Comp. ICRTC 2022*, 235–247.
- Chia, L.N., Talla Mbé, J.H., Tatietse, T.T., 2025. Direct detection and classification of *E. coli* and fecal streptococci using an optical water droplet method and convolutional neural networks. *Appl. Opt.* 64 (9), C69–C73.
- Dargan, S., Kumar, M., Ayyagari, M.R., Kumar, G., 2020. A survey of deep learning and its applications: a new paradigm to machine learning. *Arch. Comput. Methods Eng.* 27, 1071–1092.
- Dinesh, P., Vickram, A., Kalyanasundaram, P., 2024. Medical image prediction for diagnosis of breast cancer disease comparing the machine learning algorithms: SVM, KNN, logistic regression, random forest and decision tree to measure accuracy. *AIP Conf. Proc.* 2853 (1), 020140.
- Gencer, K., Gencer, G., Cizmeci, I.H., 2024. Deep learning approaches for retinal image classification: a comparative study of GoogLeNet and ResNet architectures. *Int. Sci. Vocat. Stud. J.* 8 (2), 123–128.
- He, K., Zhang, X., Ren, S., Sun, J., 2015. Delving deep into rectifiers: surpassing human-level performance on imagenet classification. *Proc. IEEE Int. Conf. Comp. Vision* 1026–1034.
- Jan, B., Farman, H., Khan, M., Imran, M., Islam, I.U., Ahmad, A., Ali, S., Jeon, G., 2019. Deep learning in big data analytics: a comparative study. *Comput. Electr. Eng.* 75, 275–287.
- Khushi, H.M.T., Masood, T., Jaffar, A., Rashid, M., Akram, S., 2023. Improved multiclass brain tumor detection via customized pre-trained efficientnetb7 model. *IEEE Access* 11, 117210–117230.
- Krizhevsky, A., Sutskever, I., Hinton, G.E., 2012. Imagenet classification with deep convolutional neural networks. *Adv. Neural Inf. Proces. Syst.* 25.
- Lekeufack, M., Fonkou, T., Pamo, T., Amougou, A., 2012. Removal of faecal bacteria and nutrients from domestic wastewater in a horizontal surface flow wetland vegetated with *Echinochloa pyramidalis*. *Afr. J. Environ. Sci. Technol.* 6 (9), 337–345.
- Li, Y., Gao, A., Liu, G., Chen, C., Zhang, Y., Kang, S., Khosravi, M., 2026. An attention-assisted machine learning system for deep microorganism image classification. *Adv. Intell. Syst.* 1–17 e202501084.
- Liedji, D.W., Talla Mbé, J.H., Kenné, G., 2022. Chaos recognition using a single nonlinear node delay-based reservoir computer. *Euro. Phys. J. B* 95 (1), 18.
- Liedji, D.W., Kenné, G., Talla Mbé, J.H., 2023. Delay-based reservoir computing using Mackey–Glass oscillator and Arduino board for edge intelligence applications. *AIP Adv.* 13 (12).
- Lu, T., Han, B., Chen, L., Yu, F., Xue, C., 2021. A generic intelligent tomato classification system for practical applications using DenseNet-201 with transfer learning. *Sci. Rep.* 11 (1), 15824.
- Muhammed, M.A.E., Ahmed, A.A., Khalid, T.A., 2017. Benchmark analysis of popular imagenet classification deep cnn architectures. *Int. Conf. Smart Technol. Smart Nat.* 2017, 902–907.
- Murugan, A., Nair, S.A.H., Kumar, K.S., 2019. Detection of skin cancer using SVM, random forest and kNN classifiers. *J. Med. Syst.* 43 (8), 269.
- Ntangmo Tsafack, H., Temgoua, E., Nono Kwekap, C.H., Saah Fokou, C., Wamba, F.R., Kenfack, S., 2022. Health risk associated with the collection, transport, and storage practices of drinking water by the population of the Dschang city, West-Cameroon. *World Water Policy* 8 (1), 31–50.
- Nurliyana, M.R., Sahdan, M.Z., Wibowo, K.M., Muslihati, A., Saim, H., Ahmad, S.A., Mansor, Z., 2018. The detection method of *Escherichia coli* in water resources: a review. *J. Phys. Conf. Ser.* 995 (1), 012065. IOP Publishing.
- Ojha, U., Adhikari, U., Singh, D.K., 2017. Image annotation using deep learning: a review. In: 2017 International Conference on Intelligent Computing and Control (I2C2), pp. 1–5.
- Planinsic, G., 2001. Water-drop projector. *Phys. Teach.* 39 (2), 76–79.
- Poomrittigul, S., Chomkwah, W., Tanpatanan, T., Sakornatanant, S., Treebupachatsakul, T., 2022. A comparison of deep learning CNN architecture models for classifying bacteria. In: 2022 37th International Technical Conference on Circuits/Systems, Computers and Communications (ITC-CSCC), pp. 290–293.
- Raja, R.V., Jayashankari, J., Sheela, S., Daisy, S.J.S., Gokilam, G.G., Joel, M.R., 2026. Metrics and techniques for evaluating machine learning models and optimization algorithms. *AI Model Design Data Manag. Dis. Predict.* 193–222.
- Santurkar, S., Tsipras, D., Ilyas, A., Madry, A., 2018. How does batch normalization help optimization? *Adv. Neural Inf. Proces. Syst.* 31.
- Shaily, T., Kala, S., 2020. Bacterial image classification using convolutional neural networks. In: 2020 IEEE 17th India Council International Conference (INDICON), pp. 1–6.
- Sharma, K., 2025. Cardiomegaly disease classification using convolutional neural networks and EfficientNet. *Vasc. Endovasc. Rev.* 8 (2), 167–180.
- Showkat, S., Qureshi, S., 2022. Efficacy of transfer learning-based ResNet models in chest X-ray image classification for detecting COVID-19 pneumonia. *Chemom. Intell. Lab. Syst.* 224, 104534.
- Simonyan, K., Zisserman, A., 2014. Very deep convolutional networks for large-scale image recognition. *arXiv Preprint arXiv:1409.1556*.

- Szegedy, C., Liu, W., Jia, Y., Sermanet, P., Reed, S., Anguelov, D., Erhan, D., Vanhoucke, V., Rabinovich, A., 2015. Going deeper with convolutions. *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.* 1–9.
- Talo, M., 2019. An automated deep learning approach for bacterial image classification. *arXiv Preprint arXiv:1912.08765*.
- Tang, J.W., Yuan, Q., Wen, X.R., Usman, M., Tay, A.C.Y., Wang, L., 2024. Label-free surface-enhanced Raman spectroscopy coupled with machine learning algorithms in pathogenic microbial identification: current trends, challenges, and perspectives. *Interdisc. Med.* 2 (3), e20230060.
- Tanko, S.M., Sani, M., Ahmad, A., 2024. Enhancing bacteria classification using image processing and convolutional neural network. *J. Basics Appl. Sci. Res.* 2 (1), 156–161.
- Wang, P., Fan, E., Wang, P., 2021. Comparative analysis of image classification algorithms based on traditional machine learning and deep learning. *Pattern Recogn. Lett.* 141, 61–67.
- Wenkack Liedji, D., Talla Mbé, J.H., Kenne, G., 2022. Classification of hyperchaotic, chaotic, and regular signals using single nonlinear node delay-based reservoir computers. *Chaos: an interdisciplinary. J. Nonlin. Sci.* 32 (12).
- Zhang, B., Gao, C., 2022. Deep residual network for image super-resolution reconstruction. In: 2022 12th International Conference on CYBER Technology in Automation, Control, and Intelligent Systems (CYBER), pp. 620–623.